ECE 121B Formula reference

01 / 08 Carrier statistics & doping

Lectures 1-12 / HW 1-3

Start with equilibrium, occupation, carrier densities, then charge neutrality.

Reference values at 300 K

L1 p25; L3 p8–10

$$q = 1.602 \times 10^{-19}\,\mathrm{C}, \quad k_B = 8.617 \times 10^{-5}\,\mathrm{eV/K}$$

(1a) For V_T, use k_B = 1.381 × 10⁻²³ J/K with q in C.

$$V_T = \frac{k_B T}{q} = 25.85\,\mathrm{mV}$$

(1b) Thermal voltage in transport and junction formulas.

$$E_{g,\mathrm{Si}} \approx 1.12\,\mathrm{eV}, \quad \varepsilon_{\mathrm{Si}} = 11.7\varepsilon_0$$

(1c) Silicon gap and permittivity; ε0 = 8.85 × 10⁻¹⁴ F/cm.

$$N_C = 2.82 \times 10^{19}, \quad N_V = 1.83 \times 10^{19}\,\mathrm{cm^{-3}}$$

(1d) Effective band-state densities at 300 K.

$$n_i \approx 10^{10}\,\mathrm{cm^{-3}}$$

(1e) Rounded course value for silicon at 300 K.

1. Occupation and carrier statistics

L2 p15, 30–32; HW1 P1–2

$$f(E) = \frac{1}{1+\exp[(E-E_F)/(k_B T)]}$$

(1f) Probability that an electron state is occupied.

$$f(E_F) = \frac{1}{2}$$

(1g) At the Fermi level, regardless of temperature.

$$f(E_F + k_B T) = \frac{1}{1+e} \approx 0.269$$

(1h) One-kT occupation check; HW1 P2.

$$f(E) \approx e^{-(E-E_F)/(k_B T)}$$

(1i) Boltzmann tail when E − E_F ≥ 3k_BT.

$$1 - f(E) \approx e^{-(E_F-E)/(k_B T)}$$

(1j) Empty-state probability below E_F.

Density of states and thermal carriers

L2 p8–10, 26; L3 p6–12

$$g_C(E) \propto \sqrt{E - E_C}, \quad g_V(E) \propto \sqrt{E_V - E}$$

(1k) Allowed-state density near each band edge.

$$n = \int_{E_C}^{\infty} g_C f\, dE, \quad p = \int_{-\infty}^{E_V} g_V (1-f)\, dE$$

(1l) Count occupied conduction and empty valence states.

$$N_C = 2\left(\frac{2\pi m^*_{n,\mathrm{DOS}} k_B T}{h^2}\right)^{3/2}$$

(1m) Conduction-band effective density of states; use DOS mass.

$$N_V = 2\left(\frac{2\pi m^*_{p,\mathrm{DOS}} k_B T}{h^2}\right)^{3/2}$$

(1n) Valence-band effective density of states; use DOS mass.

> Use kg, J and s in the DOS equations; divide m⁻³ by 10⁶ to obtain cm⁻³.

$$N_{C,V}(T) = N_{C,V}(300\,\mathrm{K})\left(\frac{T}{300\,\mathrm{K}}\right)^{3/2}$$

(1o) Scale effective state densities with T; fixed DOS mass.

$$n = N_C e^{-(E_C-E_F)/(k_B T)}, \quad p = N_V e^{-(E_F-E_V)/(k_B T)}$$

(1p) Carrier densities in the nondegenerate limit.

$$n_i = \sqrt{N_C N_V}\, e^{-E_g/(2k_B T)}$$

(1q) Intrinsic concentration; HW1 temperature and gap changes.

Units and polarity reminders

HW1 P4–5; HW2 P4

$$1\,\mu\mathrm{m} = 10^{-4}\,\mathrm{cm}, \quad 1\,\mathrm{nm} = 10^{-7}\,\mathrm{cm}$$

$$1\,\mathrm{eV}/q \equiv 1\,\mathrm{V}$$

(1r) Convert lengths before R, W or diffusion; barriers to volts.

HW1: compare intrinsic concentrations

HW1 P1; L2 p26

$$\frac{n_i(T_2)}{n_i(T_1)} = \left(\frac{T_2}{T_1}\right)^{3/2} \exp\left[-\frac{E_g}{2k_B}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)\right]$$

(1s) Use for two temperatures when E_g is held fixed.

$$\frac{n_{i,2}}{n_{i,1}} = \exp\left(\frac{E_{g,1} - E_{g,2}}{2k_B T}\right)$$

(1t) Use for a bandgap change at fixed T.

2. Doping and charge neutrality

L2 p23–26; L3 p13–20; HW1 P3, P5

> **Si donors (group V):** P, As, Sb have five valence electrons; they supply mobile electrons (n-type, N_D^+). **Si acceptors (group III):** B, Al, Ga, In have three valence electrons; they create holes (p-type, N_A^-). L2 p17–19.

$$np = n_i^2$$

(2a) Mass action at thermal equilibrium only.

$$n + N_A^- = p + N_D^+$$

(2b) Charge neutrality; use ionized dopants.

$$\Delta N = N_D - N_A$$

(2c) Net donor concentration for compensated material.

$$n \approx N_D - N_A, \quad p \approx n_i^2/n$$

(2d) Majority and minority carriers for n-type; full ionization.

$$p \approx N_A - N_D, \quad n \approx n_i^2/p$$

(2e) Majority and minority carriers for p-type; full ionization.

$$n = \frac{\Delta N + \sqrt{\Delta N^2 + 4n_i^2}}{2}$$

(2f) Exact neutrality result near intrinsic or compensated conditions.

$$p = \frac{n_i^2}{n}$$

(2g) Recover the exact hole density at equilibrium.

> HW1 P5: net doping sets n or p; total ionized impurities N_D + N_A affect mobility.

Fermi-level location

L3 p11–20; HW2 P3

$$E_F - E_i = k_B T \ln(n/n_i)$$

(2h) Find equilibrium Fermi shift from electron density.

$$E_i - E_F = k_B T \ln(p/n_i)$$

(2i) Find equilibrium Fermi shift from hole density.

$$E_C - E_F = k_B T \ln(N_C/n)$$

(2j) Conduction-edge separation in Boltzmann regime.

$$E_F - E_V = k_B T \ln(N_V/p)$$

(2k) Valence-edge separation in Boltzmann regime.

$$E_i = \frac{E_C + E_V}{2} + \frac{k_B T}{2}\ln\frac{N_V}{N_C}$$

(2l) Intrinsic level when conduction and valence DOS differ.

$$E_V + 3k_B T \lesssim E_F \lesssim E_C - 3k_B T$$

(2m) Check nondegenerate approximation before using exponentials.

> Assumptions: thermal equilibrium, parabolic bands and full ionization unless noted. At high doping use Fermi-Dirac statistics.

3. Drift, mobility and resistivity

L3 p26–41; L4 p4–22; HW1 P4–5

$$v_n = -\mu_n \mathcal{E}, \quad v_p = +\mu_p \mathcal{E}$$

(3a) Electron velocity opposes the field; hole velocity follows.

$$\mu = \frac{q\tau}{m^*}$$

(3b) Relate mobility to mean scattering time.

$$\ell = v_{\text{th}}\tau = v_{\text{th}}\frac{\mu m^*}{q}$$

(3c) Mean free path from thermal speed; HW1 P4.

$$\frac{1}{\mu} = \frac{1}{\mu_{\text{ph}}} + \frac{1}{\mu_{\text{imp}}}$$

(3d) Matthiessen rule; lower mobility dominates.

$$\mu_{\text{ph}} \propto T^{-3/2}, \quad \mu_{\text{imp}} \propto \frac{T^{3/2}}{N_D + N_A}$$

(3e) Estimate which scattering process changes with T.

$$\sigma = q(n\mu_n + p\mu_p), \quad \rho_{\text{res}} = 1/\sigma$$

(3f) Conductivity and resistivity from carrier densities.

$$R = \rho_{\text{res}} L/A$$

(3g) Convert material resistivity to device resistance.

> HW1 P5: convert dimensions to cm before using Ω·cm. Mobility follows total dopant concentration; resistivity curves assume one dopant type.

Drift and diffusion current

L3 p28–32; L4 p18–27

$$J_{n,\text{drift}} = qn\mu_n\mathcal{E}, \quad J_{p,\text{drift}} = qp\mu_p\mathcal{E}$$

(3h) Conventional current from field-driven carriers.

$$J_{n,\text{diff}} = qD_n\frac{dn}{dx}$$

(3i) Electron diffusion current from density gradient.

$$J_{p,\text{diff}} = -qD_p\frac{dp}{dx}$$

(3j) Hole diffusion current from density gradient.

$$J_n = qn\mu_n\mathcal{E} + qD_n\frac{dn}{dx}$$

(3k) Total electron current: drift plus diffusion.

$$J_p = qp\mu_p\mathcal{E} - qD_p\frac{dp}{dx}$$

(3l) Total hole current: drift plus diffusion.

$$\frac{D_n}{\mu_n} = \frac{D_p}{\mu_p} = V_T$$

(3m) Einstein relation for nondegenerate carriers.

$$\mathcal{E} = -\frac{d\psi}{dx} = \frac{1}{q}\frac{dE_C}{dx}$$

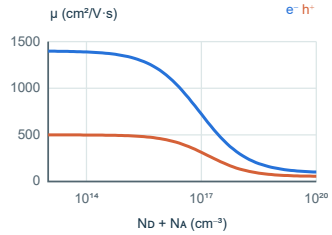
(3n) Electric field from potential or band-edge slope.

$$\mathcal{E} = -\frac{V_T}{n}\frac{dn}{dx}$$

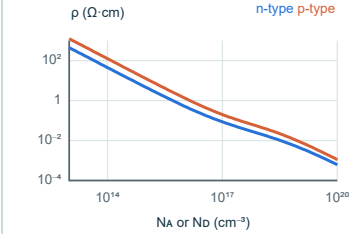
(3o) Equilibrium field that cancels electron diffusion.

> For the band-edge slope, E_C must be in joules when q is in coulombs.

Si mobility · 300 K (L3 p36)



Si resistivity · 300 K (L3 p40)



4. Continuity and generation

L4 p29–32; HW2 P1–2

$$\frac{\partial n}{\partial t} = \frac{1}{q}\frac{\partial J_n}{\partial x} + G - R$$

(4a) Electron balance in one dimension.

$$\frac{\partial p}{\partial t} = -\frac{1}{q}\frac{\partial J_p}{\partial x} + G - R$$

(4b) Hole balance in one dimension.

$$\frac{\partial n}{\partial t} = \mu_n\frac{\partial(n\mathcal{E})}{\partial x} + D_n\frac{\partial^2 n}{\partial x^2} + G - R$$

(4c) Expanded electron form: drift, diffusion, generation and loss.

$$\frac{\partial p}{\partial t} = -\mu_p\frac{\partial(p\mathcal{E})}{\partial x} + D_p\frac{\partial^2 p}{\partial x^2} + G - R$$

(4d) Expanded hole form for constant mobility and diffusivity.

$$\Delta n = \Delta p$$

(4e) Pair generation changes both carrier populations equally.

$$\Delta n = G_L\tau_n$$

(4f) Uniform steady low-injection photogeneration; HW2 P1.

$$n = n_0 + \Delta n, \quad p = p_0 + \Delta p$$

(4g) Total density during illumination.

$$\frac{\partial \Delta p}{\partial t} = D_p\frac{\partial^2 \Delta p}{\partial x^2} - \frac{\Delta p}{\tau_p} + G_L$$

(4h) Field-free low-injection excess-hole generation equation.

$$\frac{\partial \Delta n}{\partial t} = D_n\frac{\partial^2 \Delta n}{\partial x^2} - \frac{\Delta n}{\tau_n} + G_L$$

(4i) Electron counterpart for a p-type neutral region.

> Expanded forms assume constant μ and D. Low injection: excess carriers ≪ equilibrium majority density; in p-type Si compare Δn with p₀.

Minority diffusion and illuminated bar

L4 p29–32; L5 p5–17; HW2 P2

$$D_p\frac{d^2\Delta p}{dx^2} - \frac{\Delta p}{\tau_p} = 0$$

(4j) Steady, field-free, unilluminated interior.

$$L_p = \sqrt{D_p\tau_p}, \quad L_n = \sqrt{D_n\tau_n}$$

(4k) Distance excess minority carriers survive.

$$\Delta p(x) = Ae^{-x/L_p} + Be^{x/L_p}$$

(4l) General one-dimensional diffusion solution.

$$\Delta p(0) = \Delta p(L) = \gamma N_D$$

(4m) Boundary condition for equal end illumination.

$$\Delta p(x) = \gamma N_D\frac{\cosh[(x - L/2)/L_p]}{\cosh[L/(2L_p)]}$$

(4n) Symmetric bar profile satisfying both ends.

$$J_p(x) = -qD_p\frac{d\Delta p}{dx}$$

(4o) Hole current at an edge; differentiate the profile.

$$\Delta p(x) = \Delta p(0)e^{-x/L_p}$$

(4p) Single-edge decay into a long neutral region.

$$\Delta p(t) = \Delta p(0)e^{-t/\tau_p}$$

(4q) Time decay after generation stops; no spatial current.

$$J_p(0) = \frac{qD_p\gamma N_D}{L_p}\tanh\frac{L}{2L_p}$$

(4r) Hole diffusion current entering the equally lit bar.

> In HW2 P2, γ is stated as 10^{-4}. Use the stated boundary value when checking the worked solution.

5. Quasi-Fermi levels

L5 p21–23; HW2 P3

$$n = n_i e^{(F_n - E_i)/(k_B T)}$$

(5a) Electron density out of equilibrium.

$$p = n_i e^{(E_i - F_p)/(k_B T)}$$

(5b) Hole density out of equilibrium.

$$F_n = E_i + k_B T \ln(n/n_i)$$

(5c) Locate electron quasi-Fermi level.

$$F_p = E_i - k_B T \ln(p/n_i)$$

(5d) Locate hole quasi-Fermi level.

$$np = n_i^2 e^{(F_n - F_p)/(k_B T)}$$

(5e) Connect quasi-Fermi splitting to pair density.

> At equilibrium F_n = F_p = E_F. HW2 P3: low injection barely moves the majority quasi-Fermi level; high injection moves both.

Electrostatic potential and Poisson equation

L5 p24–25; L6 p17–20

$$\rho_{\text{ch}} = q(p - n + N_D^+ - N_A^-)$$

(5f) Local space charge, including ionized dopants.

$$\frac{d\mathcal{E}}{dx} = \frac{\rho_{\text{ch}}}{\varepsilon_s}$$

(5g) Field slope from charge density.

$$\mathcal{E} = -\frac{d\psi}{dx}, \qquad \frac{d^2\psi}{dx^2} = -\frac{\rho_{\text{ch}}}{\varepsilon_s}$$

(5h) Potential slope and curvature.

$$\psi(x_2) - \psi(x_1) = V_T \ln\frac{n(x_2)}{n(x_1)}$$

(5i) Equilibrium potential difference from electron densities.

6. Abrupt PN junction: voltage and width

L6 p17–29

$$V_{bi} = V_T \ln\left(\frac{N_A N_D}{n_i^2}\right)$$

(6a) Built-in potential at equilibrium.

One degenerately doped PN side

L6 p18

$$qV_{bi} = (E_i - E_F)_p + (E_F - E_i)_n$$

(6b) Add the p-side and n-side Fermi-level offsets.

$$p^+n: \quad qV_{bi} \approx \frac{E_g}{2} + k_B T \ln\frac{N_D}{n_i}$$

$$n^+p: \quad qV_{bi} \approx \frac{E_g}{2} + k_B T \ln\frac{N_A}{n_i}$$

(6c) Band-edge Fermi levels, midgap E_i; light side nondegenerate.

$$U = V_{bi} - V_A > 0$$

(6d) Voltage across depletion; forward PN bias lowers U.

$$W = \sqrt{\frac{2\varepsilon_s U}{q}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)}$$

(6e) Total width; one-sided: $W \approx \sqrt{2\varepsilon_s U/(qN_{\text{light}})}$.

$$x_n = \sqrt{\frac{2\varepsilon_s V_{bi}}{q}\frac{N_A}{N_D(N_A + N_D)}}$$

$$x_p = \sqrt{\frac{2\varepsilon_s V_{bi}}{q}\frac{N_D}{N_A(N_A + N_D)}}$$

(6f) At equilibrium (V_A = 0). Under bias, replace V_bi with U = V_bi – V_A.

Abrupt PN junction: field and charge

L6 p21–29

$$N_A x_p = N_D x_n, \quad W = x_p + x_n$$

(6g) Charge neutrality and width split.

$$x_p = \frac{W N_D}{N_A + N_D}, \quad x_n = \frac{W N_A}{N_A + N_D}$$

(6h) Find depletion distance on each side.

$$|\mathcal{E}_{\max}| = \frac{qN_A x_p}{\varepsilon_s} = \frac{qN_D x_n}{\varepsilon_s} = \frac{2U}{W}$$

(6i) Peak triangular field; voltage is field area.

$$|Q| = qAN_A x_p = qAN_D x_n$$

(6j) Magnitude of charge on either side.

$$\mathcal{E}_p(x) = -\frac{qN_A}{\varepsilon_s}(x + x_p)$$

(6k) Field on the depleted p side: –x_p ≤ x ≤ 0.

$$\mathcal{E}_n(x) = \frac{qN_D}{\varepsilon_s}(x - x_n)$$

(6l) Field on the depleted n side: 0 ≤ x ≤ x_n.

$$U = -\int_{-x_p}^{x_n} \mathcal{E}(x)\,dx = \frac{|\mathcal{E}_{\max}|W}{2}$$

(6m) Voltage equals the triangular field area.

7. Depletion capacitance and C–V

L6 p9–11

$$C = \frac{\varepsilon_s A}{W}$$

(7a) Small-signal depletion capacitance.

$$i_{\text{ac}} = C\frac{dv_{\text{ac}}}{dt}$$

(7b) AC displacement current through depletion.

$$\frac{1}{C^2} = \frac{2(V_{bi} - V_A)}{q\varepsilon_s A^2 N}$$

(7c) Linear C^{-2} plot for a one-sided junction.

$$m = \frac{d(1/C^2)}{dV_A} = -\frac{2}{q\varepsilon_s A^2 N}$$

(7d) Slope yields the lightly doped density.

$$N = -\frac{2}{q\varepsilon_s A^2 m}$$

(7e) Extract doping from measured C–V slope.

> These C–V signs use the n-type forward-bias convention. The voltage-axis intercept is V_bi.

$$qV_{bi} = \Phi_{Bn} - (E_C - E_F)_{\text{FB}}$$
$$= \Phi_{Bn} - k_B T \ln\left(\frac{N_C}{N_D}\right)$$

(7f) For an n-type Schottky contact, use V_bi from the C–V intercept and N_D from its slope to find Φ_Bn; energies in eV.

Check before substitution

Lectures 5–6

> Use ε in F/cm, area in cm² and dopants in cm⁻³. Inside W, fixed dopant charge dominates. Reverse bias widens W and lowers C; mark p and n sides before assigning x_p and x_n.

Continuity: useful limiting cases

L4 p29–32; HW2 P1–2

> **Zero field / no drift: (4s)** $\mathcal{E} = 0 \Rightarrow J_n = qD_n n'$, $J_p = -qD_p p'$. Diffusion can still flow.
>
> **Thermal equilibrium: (4t)** $\partial_t n = \partial_t p = 0$, $G = R$, $J_n = J_p = 0$. Drift and diffusion may cancel.
>
> **Uniform, steady light: (4u)** $0 = G_L - \Delta n/\tau_n \Rightarrow \Delta n = G_L \tau_n$ for minority electrons in p-type Si.

8. Metal–semiconductor contacts

L5 p27–39; L6 p4–8; HW2 P4

$$\Phi_{Bn} = \Phi_M - \chi, \quad \Phi_{Bp} = E_g - \Phi_{Bn}$$

(8a) Ideal barrier energies; Φ_M is metal work function, χ electron affinity.

$$\Phi_{Bn} = qV_{bi,n} + (E_C - E_F)$$

(8b) Barrier = band bending + bulk separation; every term is an energy (eV).

$$E_C - E_F = k_B T \ln(N_C/N_D)$$

(8c) Bulk n-type band-edge separation.

$$E_F - E_V = k_B T \ln(N_V/N_A)$$

(8d) Bulk p-type band-edge separation.

$$V_{bi,n} = \frac{\Phi_{Bn} - (E_C - E_F)}{q}$$

(8e) n-type built-in voltage from barrier and bulk Fermi level.

$$V_{bi,p} = \frac{\Phi_{Bp} - (E_F - E_V)}{q}$$

(8f) p-type built-in voltage from barrier and bulk Fermi level.

$$W_n = \sqrt{\frac{2\varepsilon_s(V_{bi,n} - V_A)}{qN_D}}$$

(8g) n-type contact: reverse bias V_A < 0 widens W.

$$W_p = \sqrt{\frac{2\varepsilon_s(V_{bi,p} + V_A)}{qN_A}}$$

(8h) p-type polarity used in L6: V_A > 0 is reverse bias.

> HW2 P4 realistic NiSi/Si values: electron barrier Φ_Bn = 0.65 eV; hole barrier Φ_Bp = 0.47 eV. Use these instead of ideal work-function estimates.

> At fixed metal and semiconductor, applied bias changes band bending and W, not the interface barrier height.

$$\frac{W_n(V_A)}{W_n(0)} = \sqrt{1 - \frac{V_A}{V_{bi,n}}}$$

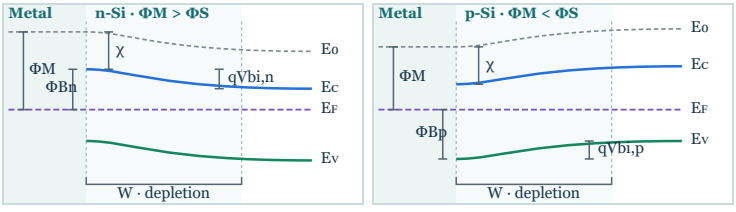
(8i) Compare n-type contact widths across biases; HW2 P4.

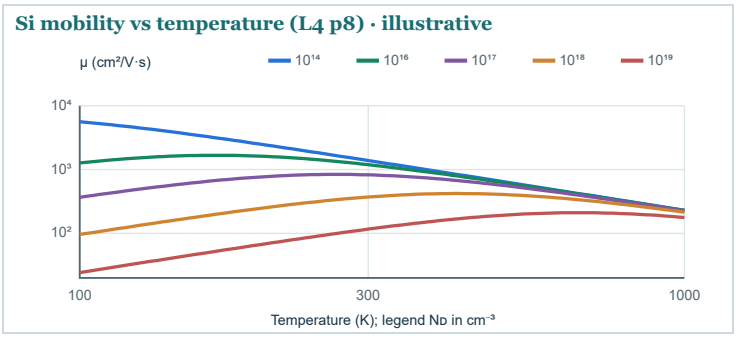
$$|Q_s| = qN_D A W_n$$

(8j) Depletion charge magnitude in an n-type contact.

Schottky barriers at equilibrium

L5 p29, 38; ideal rectifying contacts; schematic, not to scale





9. Ideal PN diode

L7 p6–20

$$n_{p0} = n_i^2/N_A, \quad p_{n0} = n_i^2/N_D$$

(9a) Minority equilibrium densities at each neutral side.

$$n_p(-x_p) = n_{p0}e^{V_A/V_T}, \quad p_n(x_n) = p_{n0}e^{V_A/V_T}$$

(9b) Minority boundary densities under applied bias.

$$J = J_0\left(e^{V_A/V_T} - 1\right), \quad I = AJ$$

(9c) Ideal current density and terminal current.

$$J(V_A \ll -V_T) \approx -J_0$$

(9d) Reverse saturation before breakdown.

$$J_0 = qn_i^2\left(\frac{D_n}{L_n N_A} + \frac{D_p}{L_p N_D}\right)$$

(9e) Reverse saturation density for long neutral regions.

$$\Delta p_n(x_n + x) = p_{n0}(e^{V_A/V_T} - 1)e^{-x/L_p}$$

(9f) Injected hole profile in the neutral n-side.

$$\Delta V_{\text{decade}} = V_T \ln 10 \approx 59.5\,\text{mV}$$

(9g) Ideal 300 K semilog forward-current slope.

$$r_d \approx \frac{V_T}{I}$$

(9h) Small-signal resistance at strong ideal forward bias.

> Ideal law assumes low injection, steady state, long neutral regions and negligible depletion-region recombination.

10. Nonideal current and breakdown

L8 p4–16

$$I_{\text{R–G}} \approx -\frac{qAn_iW}{2\tau_0}$$

(10a) Reverse generation current in depletion.

$$\tau_0 = \frac{\tau_p n_1 + \tau_n p_1}{2n_i}$$

(10b) Effective trap lifetime used in reverse generation.

$$R_{\text{net}} = \frac{np - n_i^2}{\tau_p(n + n_1) + \tau_n(p + p_1)}$$

(10c) Trap-assisted recombination; sign shows net R or G.

$$n_1 = n_i e^{(E_T - E_i)/(k_B T)}, \quad p_1 = n_i e^{(E_i - E_T)/(k_B T)}$$

(10d) Trap reference densities from trap energy E_T.

$$I_{\text{F,R–G}} \propto n_i W e^{V_A/(2V_T)}$$

(10e) Forward depletion recombination; ideality near two.

$$V_A = V_J + IR_S$$

(10f) Subtract series drop to find junction voltage.

$$V_{\text{BR}} \approx \frac{\varepsilon_s \mathcal{E}_{\text{CR}}^2}{2qN_{\text{light}}} - V_{bi}$$

(10g) One-sided critical-field estimate; not a current law.

> Avalanche: impact ionization, usually wider depletion region. Zener: tunneling through very narrow, heavily doped depletion region.

Last checks

Course conventions

> State assumptions, use consistent cm–F–V units, keep electrons and holes distinct, and check the sign of drift versus diffusion current.

11. Biased PN minority carriers

L9 p4, 9; HW3 P2, P4

Equilibrium minority densities (9a); diffusion lengths (4k).

$$\Delta n_p(-x_p) = n_{p0}(e^{V_A/V_T} - 1)$$

(11a) Excess electron density at the p-side edge.

$$\Delta p_n(x_n) = p_{n0}(e^{V_A/V_T} - 1)$$

(11b) Excess hole density at the n-side edge.

$$\Delta n_p(x) = \Delta n_p(-x_p)e^{(x+x_p)/L_n}$$

(11c) Electron profile for x ≤ −x_p in a long p base.

Hole profile in the long n base: (9f).

HW3: minority current components

HW3 P2; L7 p16

Use Einstein relation (3m), ideal diode law (9c), and saturation density (9e).

$$J_n(-x_p) = qD_n\Delta n_p(-x_p)/L_n$$

(11d) Electron injection from n into p.

$$J_p(x_n) = qD_p\Delta p_n(x_n)/L_p$$

(11e) Hole injection from p into n.

$$J_n(x) = J_n(-x_p)e^{(x+x_p)/L_n}$$

(11f) Electron minority current decays into the p region.

$$J_p(x) = J_p(x_n)e^{-(x-x_n)/L_p}$$

(11g) Hole minority current decays into the n region.

$$J_n + J_p = J_{\text{total}}$$

(11h) Current continuity: the majority part makes up the loss.

$$\frac{J_p(x_n)}{J_n(-x_p)} = \frac{D_pN_AL_n}{D_nN_DL_p}$$

(11i) Compare hole and electron injection.

$$J_p(x) = \tfrac{1}{2}J_{\text{total}}$$

(11j) Equal-current plane; solve for distance from x_n.

$$x - x_n = L_p \ln\frac{2J_p(x_n)}{J_{\text{total}}}$$

(11k) HW3 P2 equal-current distance in the n base.

> HW3 P2: use L_n in the electron exponent and L_p in the hole exponent. A printed intermediate line swaps the symbol.

12. Stored excess charge

L9 p4–7; HW3 P4

$$Q_p = qA\int_{x_n}^{\infty}\Delta p_n(x)\,dx$$

(12a) Integrate injected holes.

$$Q_p = qA\Delta p_n(x_n)L_p$$

(12b) Positive hole charge stored in the long n base.

$$Q_n = -qA\Delta n_p(-x_p)L_n$$

(12c) Signed electron charge stored in the long p base.

$$I_p = Q_p/\tau_p, \quad I_n = |Q_n|/\tau_n$$

(12d) Steady injection replaces recombined charge.

> HW3 P4: A = 100 μm² = 10⁻⁶ cm². The electron charge Q_n is negative; use its magnitude in the current formula.

13. Charge control: current → stored charge

L9 p5–7, 16–23

$$I = \frac{dQ}{dt} + \frac{Q}{\tau_p}$$

(13a) Here I is the hole-injection component in one long base; negative I extracts charge.

Long base: far-edge current vanishes, so the edge hole current is I in (13a).

$$Q(t) = I\tau_p + [Q(t_0) - I\tau_p]e^{-(t-t_0)/\tau_p}$$

(13b) General constant-current solution; set the initial charge at to.

$$Q(0) = Q_{ss} = I_F\tau_p$$

(13c) HW4 P1: initial charge after steady forward current.

$$\frac{dQ}{dt} = -I_R - \frac{Q}{\tau_p}$$

(13d) Reverse current extracts stored charge during the storage interval.

$$Q(t) = \tau_p[(I_F + I_R)e^{-t/\tau_p} - I_R]$$

(13e) HW4 P1: reverse extraction from Q(0) = I_Fτp.

$$t_s = \tau_p \ln\left(1 + \frac{I_F}{I_R}\right)$$

(13f) Set Q(ts) = 0; for equal 1 mA currents and τp = 1 μs, ts = 0.693 μs.

Forward reapplication: put I = I_F into the general solution (13b).

$$Q(t) \simeq I_F\tau_p(1 - e^{-(t-t_0)/\tau_p})$$

(13g) Use only when Q(t0) is negligible.

Stored charge → diode voltage

L9 p16–23; quasi-static

$$Q_0 = qAp_{n0}L_p = \tau_pI_S$$

(13h) One-sided saturation current I_S; Q0 is not initial excess charge.

$$Q = Q_0(e^{v_a/V_T} - 1)$$

(13i) Boundary injection sets excess charge in a long n base.

$$v_a = V_T \ln\left(1 + \frac{Q}{Q_0}\right)$$

(13j) Convert stored excess charge to voltage.

HW4 P1 voltage: combine turn-on charge (13g), charge scale (13h), and voltage conversion (13j); t0 = 2 μs.

> Use this charge-to-voltage step only while the long-base, low-injection, quasi-static model applies. It does not describe the entire turn-off waveform after storage ends.

14. Small-signal diode around DC bias

L9 p8–12, 15; HW3 P5

$$g_d = \left.\frac{dI}{dV}\right|_{V_{DC}} \approx \frac{I_{DC}}{V_T}$$

(14a) Differential conductance at strong ideal forward bias.

Differential resistance (9h); depletion capacitance from width (7a).

$$g_d = I_0 e^{V_{DC}/V_T}/V_T$$

(14b) Exact ideal conductance before I ≫ I0.

$$C_D = \frac{dQ_{\text{stored}}}{dV} \approx \frac{\tau_n I_n + \tau_p I_p}{V_T}$$

(14c) Diffusion capacitance from stored minority charge.

$$C_J = \left|\frac{dQ_{\text{dep}}}{dV_A}\right|$$

(14d) Depletion charge response; use width form (7a).

$$C_{\text{total}} = C_J + C_D$$

(14e) Both charge responses are parallel.

$$\tau_{\text{eff}} = \frac{\tau_n I_n + \tau_p I_p}{I_n + I_p}$$

(14f) Current-weighted minority lifetime for C_D.

$$i_{\text{ac}} = g_d v_{\text{ac}} + C_{\text{total}} \frac{dv_{\text{ac}}}{dt}$$

(14g) Parallel resistor–capacitor current relation.

> HW3 P5: the supplied 1 µs / 2.95 × 10⁵ Ω quotient is about 3.4 pF, not 3.9 pF; diffusion capacitance dominates the ~0.06 pF junction term.

HW3: check built-in voltage

HW3 P1, P5

Start from built-in voltage (6a) and depletion width split (6g).

$$\mathcal{E}(0) = -qN_D x_n/\varepsilon_s$$

(14h) Peak signed field for p on left and n on right.

$$\psi(0) - \psi(-x_p) = qN_A x_p^2/(2\varepsilon_s)$$

(14i) Potential rise from p edge to junction.

> HW3 checks: P5 gives V_bi ≈ 0.84 V, W(0.6 V) ≈ 0.18 µm, C_J ≈ 0.06 pF, C_D ≈ 3.4 pF. P1 prints an extra 1/2 in peak field and uses N_D with x_p² in potential.

15. PN and Schottky depletion C–V

L9 p10–14; HW3 P3, P5

Two-sided width (6e), ionized charge (6j), junction C (7a), and C–V line, slope, and doping (7c), (7d), (7e).

$$W \simeq \sqrt{\frac{2\varepsilon_s(V_{bi} - V_A)}{qN_{\text{light}}}}$$

(15a) One-sided abrupt depletion width; forward-bias convention.

$$(1/C_J^2)_{V_A = V_{bi}} = 0$$

(15b) Extrapolated x-intercept gives V_bi.

Schottky contact: interpret the C–V line

HW3 P3; L5 p27–39

Use barriers (8a), band bending (8b), and bulk Fermi offset (8c). Recover the barrier by combining (8b) and (8c).

> For equal contact areas, the steeper 1/C² slope has lower N_D. Different doping shifts the bulk Fermi offset, so intercepts alone do not give an exact metal work-function difference.

Quick selection by operating regime

L9 p10–20

$$V_A < 0: \; C_{\text{total}} \approx C_J$$

(15c) Reverse bias: little injected minority charge.

$$V_A \gg V_T: \; C_D \approx \tau_{\text{eff}} I/V_T$$

(15d) Strong forward bias: diffusion storage can dominate.

$$\omega_{RC} = 1/(r_d C_{\text{total}})$$

(15e) Local small-signal RC corner estimate.

> Track A in cm², ε_s in F/cm, N in cm⁻³ and W in cm. An inverse-square C–V intercept is an extrapolation, not a forward-bias measurement.

16. PN junction as temperature sensor

L10 p7

$$I = I_S(T)\left(e^{V_F/(\eta V_T)} - 1\right)$$

(16a) Forward diode current; η is ideality factor.

$$V_F = \eta V_T \ln\left(1 + \frac{I}{I_S(T)}\right)$$

(16b) At fixed current, read temperature from voltage.

$$I_S(T) \propto T^3 e^{-E_g/(k_B T)}$$

(16c) Illustrative silicon temperature trend when transport factors are fixed.

$$S_T = \left.\frac{dV_F}{dT}\right|_I$$

(16d) Calibrated voltage sensitivity at fixed current.

> At ordinary bias a silicon diode usually needs less V_F as T rises. Calibrate at a fixed current; I_S(T) and series resistance are device-dependent.

17. Varactor: reverse-bias tuning

L10 p8

$$C_j(V_r) = C_0\left(1 + \frac{V_r}{V_{bi}}\right)^{-n}$$

(17a) Stronger reverse bias lowers C; for V_r ≫ V_bi, C_j scales as V_r^{-n}.

$$n = \frac{1}{m+2}$$

(17b) Graded profile N(x) ∝ x^m.

$$m = 1 \Rightarrow n = \tfrac{1}{3}, \quad m = 0 \Rightarrow n = \tfrac{1}{2}$$

(17c) Linear grade versus abrupt junction.

$$m = -\tfrac{3}{2} \Rightarrow n = 2$$

(17d) Hyperabrupt junction: stronger tuning.

$$\omega_0 = \frac{1}{\sqrt{LC_j}}, \quad f_0 = \frac{1}{2\pi\sqrt{LC_j}}$$

(17e) LC oscillator frequency rises as C_j falls.

$$\frac{f_0(V_r)}{f_0(0)} = \left(1 + \frac{V_r}{V_{bi}}\right)^{n/2}$$

(17f) Frequency tuning ratio at fixed inductance.

18. Photodiode collection

L10 p9–10, 13; HW4 P2

Dark diode current: use ideal diode law (9c) with I = AJ.

$$E_\gamma = h\nu = hc/\lambda$$

(18a) Photon energy and wavelength.

$$\lambda(\mu\text{m}) \approx \frac{1.24}{E_\gamma(\text{eV})}$$

(18b) Quick optical wavelength conversion.

$$E_\gamma \geq E_g$$

(18c) Photon must meet the gap for band-to-band absorption.

$$\lambda_c(\mu\text{m}) \approx 1.24/E_g(\text{eV})$$

(18d) Approximate absorption cutoff wavelength.

$$I_L \approx -qA(L_p + W + L_n)G_L$$

(18e) Uniform-generation collection estimate; sign opposes diode current.

$$I_{\text{light}} = I_S(e^{V_A/V_T} - 1) - I_{\text{ph}}$$

(18f) Photocurrent opposes dark current.

> Photocarriers formed in depletion are field swept. In neutral regions, minority carriers generally need to diffuse to the junction before recombining.

19. Solar-cell I–V and open circuit

L10 p10–12

Use illuminated diode current (18f), with I₀ = I_S; open circuit sets terminal I = 0.

$$V_{OC} = V_T \ln\left(1 + \frac{I_{\text{ph}}}{I_0}\right)$$

(19a) At I = 0, forward diode current balances photocurrent.

$$I(0) = -I_{\text{ph}}, \quad I_{SC} = I_{\text{ph}}$$

(19b) Signed terminal current and short-circuit magnitude.

$$P_{\text{out}} = -VI(V)$$

(19c) Electrical power in the generating quadrant.

$$FF = \frac{P_{\max}}{V_{OC} I_{SC}}$$

(19d) Fill factor: maximum power versus ideal rectangle.

$$\left.\frac{dP_{\text{out}}}{dV}\right|_{V_{\text{MP}}} = 0$$

(19e) Maximum-power point condition.

> Use current density J throughout if area is unknown; then output power is per area. Series and shunt losses lower a real cell's fill factor.

20. LED emission

L10 p14

$$E_\gamma \approx E_g$$

(20a) Direct-gap radiative recombination sets photon energy.

LED wavelength: combine band-gap photon energy (20a) with the wavelength conversion (18b).

$$R_\gamma \approx \eta_{\text{rad}} \frac{I}{q}$$

(20b) Internal photon-generation rate from radiative fraction.

$$\eta_{\text{ext}} = \eta_{\text{rad}} \eta_{\text{extract}}$$

(20c) External output also needs photon escape.

> Visible anchors: red ~650 nm, green ~570 nm, blue ~450 nm. Direct-band-gap materials emit efficiently; extraction losses reduce external output.

HW4 P2: minority-hole continuity

L4 continuity; L10 photodiode

$$\frac{\partial \Delta p_n}{\partial t} = D_p \frac{\partial^2 \Delta p_n}{\partial x^2} - \frac{\Delta p_n}{\tau_p} + G_L$$

(18g) Generation adds holes; diffusion and recombination redistribute or remove them.

Set the time derivative to zero for the steady-state minority-hole equation (4i).

$$\Delta p_n(\infty) = G_L \tau_p$$

(18h) Far from the junction, uniform generation balances recombination.

ECE 121B Formula reference

08 / 08 MOS capacitor

Compare p-type and n-type substrates; follow band bending, charge, and C-V.

21. MOS stack and flat band

L11 p4–11; L12 p4

$$C'_{ox} = \frac{\varepsilon_{ox}}{t_{ox}}, \quad C_{ox} = AC'_{ox}$$

(21a) Oxide capacitance per area and total capacitance.

$$\varepsilon_{ox} \approx 3.9\varepsilon_0$$

(21b) Silicon dioxide permittivity.

$$V_{FB} = \frac{\Phi_M - \Phi_S}{q} - \frac{Q'_{ox}}{C'_{ox}}$$

(21c) Flat-band voltage; HW4 uses Qox = 0, so only the work-function term remains.

$$V_G = V_{FB} + V_{ox} + \psi_s$$

(21d) Gate bias divides between oxide and semiconductor surface potential.

$$V_{ox} = -Q'_s / C'_{ox}$$

(21e) Oxide drop from semiconductor sheet charge.

$$E_C(0) - E_C(\infty) = -q\psi_s$$

(21f) Surface band bending for electrostatic potential ψ_s.

> Primes denote charge or capacitance per area. Work-function differences in eV are numerically volts after division by q. At flat band, ψ_s = 0.

22. P-type accumulation and depletion

L11 p12; L12 p5–13

$$V_G < V_{FB} \Rightarrow \text{hole accumulation}$$

(22a) Negative gate relative to flat band attracts holes.

$$V_{FB} < V_G < V_T^{\text{MOS}} \Rightarrow \text{depletion}$$

(22b) Positive gate repels holes and exposes acceptors.

$$W = \sqrt{\frac{2\varepsilon_{Si}\psi_s}{qN_A}}$$

(22c) Depletion width from Poisson's equation.

$$Q'_{dep} = -qN_A W = -\sqrt{2q\varepsilon_{Si}N_A\psi_s}$$

(22d) Negative fixed-ion charge per area in p-type Si.

HW4 P3: N-type MOS and band bending

L11–12; ideal oxide

For the bulk n-type band-edge offset use (8c); field from potential is (5h).

$$\Phi = E_{\text{vac}} - E_F$$

(22e) Work function is vacuum level minus Fermi level.

$$\Phi_S = \chi + (E_C - E_F)$$

(22f) N-type semiconductor work function.

$$\Phi_S \simeq 4.136\,\text{eV}, \quad V_{FB} \simeq +1.06\,\text{V}$$

(22g) HW4 values; an energy difference in eV is numerically volts after division by q.

$$\phi_F = V_T \ln(N_D/n_i), \quad \psi_{s,inv} \simeq -2\phi_F$$

(22h) N-type strong inversion; HW4 gives ψs ≈ −0.954 V.

$$E_{C,V}(x) = E_{C,V,\text{ref}} - q\phi(x)$$

(22i) Both band edges move together with electrostatic potential.

$$E_{F,M} - E_{F,S} = -qV_G$$

(22j) Gate-to-semiconductor Fermi-level offset.

$$V_G = V_{FB}, \quad \psi_s = V_{ox} = 0$$

(22k) Flat-band condition.

> N-type Si: accumulation bends both bands down; depletion and inversion bend them up. The Fermi level is flat within each material at equilibrium.

23. Strong inversion and threshold

L12 p14–21

$$\phi_F = V_T \ln(N_A/n_i)$$

(23a) Positive bulk Fermi-potential magnitude for p-type Si.

$$\psi_{s,T} = 2\phi_F$$

(23b) Strong-inversion threshold surface potential.

$$n_s \approx (n_i^2/N_A)e^{\psi_s/V_T}$$

(23c) Surface electrons rise exponentially with band bending.

$$n_s(2\phi_F) \approx N_A$$

(23d) Strong-inversion surface criterion.

$$W_T = \sqrt{\frac{4\varepsilon_{Si}\phi_F}{qN_A}}$$

(23e) Approximate maximum depletion width at threshold.

$$Q'_{dep,T} = -\sqrt{4q\varepsilon_{Si}N_A\phi_F}$$

(23f) Threshold depletion sheet charge.

$$V_T^{\text{MOS}} = V_{FB} + 2\phi_F + \frac{\sqrt{4q\varepsilon_{Si}N_A\phi_F}}{C'_{ox}}$$

(23g) Threshold voltage for ideal p-type substrate.

$$Q'_{inv} \approx -C'_{ox}(V_G - V_T^{\text{MOS}})$$

(23h) Electron inversion charge above threshold.

> Here V_T without a MOS superscript is thermal voltage; V_T^MOS means transistor threshold. Added gate bias above threshold mainly increases inversion charge.

24. MOS capacitance versus frequency

L12 p22–29

For depletion capacitance per area use (7a) with A = 1; AC current uses (14g).

$$C' = |dQ'_G/dV_G|$$

(24a) Small-signal gate-charge slope.

$$\frac{1}{C'} = \frac{1}{C'_{ox}} + \frac{1}{C'_{dep}}$$

(24b) Oxide and depletion regions act in series.

$$C'_{min} = \left(\frac{1}{C'_{ox}} + \frac{W_T}{\varepsilon_{Si}}\right)^{-1}$$

(24c) High-frequency inversion limit: depletion stays near W_T.

$$L_D = \sqrt{\varepsilon_{Si}V_T/(qN_A)}$$

(24d) Extrinsic Debye screening length at flat band.

$$C'_{FB} = \left(\frac{1}{C'_{ox}} + \frac{L_D}{\varepsilon_{Si}}\right)^{-1}$$

(24e) Flat-band capacitance below C_ox due to finite screening.

$$C'_{LF,inv}, \; C'_{acc} \to C'_{ox}$$

(24f) Minority carriers follow low-frequency inversion; accumulation charge sits near oxide.

> For n-type Si reverse the gate-voltage polarities and majority/minority carriers. Low- versus high-frequency inversion depends on electron or hole supply, not just DC bias.